Example 1.4 Without using truth tables, prove the following:

(i) $(\neg p \vee q) \wedge (p \wedge (p \wedge q) \equiv p \wedge q$

(ii) $p \to (q \to p) \equiv \neg p \to (p \to q)$

(iii) $\neg(p \leftrightarrow q) \equiv (p \vee q) \wedge \neg(p \vee q) \equiv (p \wedge \neg q) \vee (\neg p \wedge q)$

(i) $(\neg p \vee q) \wedge (p \wedge (p \wedge q)) \equiv (\neg p \vee q) \wedge (p \wedge p) \wedge q$, by associative law
$\equiv (\neg p \vee q) \wedge (p \wedge q)$, by idempotent law
$\equiv (p \wedge q) \wedge (\neg p \vee q)$, by commutative law
$\equiv ((p \wedge q) \wedge \neg p) \vee ((p \wedge q) \wedge q)$, by distributive law
$\equiv (\neg p \wedge (p \wedge q)) \vee ((p \wedge q) \wedge q)$, by commutative law
$\equiv ((\neg p \wedge p) \wedge q) \vee (p \wedge (q \wedge q)$, by associative law
$\equiv (F \vee q) \vee (p \wedge q)$, by complement and idempotent law
$\equiv F \vee (p \wedge q)$, by dominant law
$\equiv p \wedge q$, by dominant law.

(ii) $p \to (q \to p) \equiv \neg p \vee (q \to p)$ [Refer to Table 1.10]
$\equiv \neg p \vee (\neg q \vee p)$ [Refer to Table 1.10]
$\equiv \neg q \vee (p \vee \neg p)$, by commutative and associative laws
$\equiv \neg p \vee \mathrm{T}$, by complement law
$\equiv \mathrm{T}$, by dominant law (1)

$\neg p \to (p \to q) \equiv p \vee (p \to q)$, by (1) of Table 1.10
$\equiv p \vee (\neg p \vee q)$, by (1) of Table 1.10
$\equiv (p \vee \neg p) \vee q$, by associative law
$\equiv T \vee q$, by complement law
$\equiv T$, by dominant law, (2)

From (1) and (2), the result follows.

(iii) $\neg(p \leftrightarrow q) \equiv \neg((p \to q) \wedge (q \to p))$, from Table 1.11
$\equiv \neg((\neg p \vee q) \wedge (\neg q \vee q))$, from Table 1.10
$\equiv \neg[((\neg p \vee q) \wedge \neg q) \vee ((\neg p \vee q) \wedge p]$, by distributive law
$\equiv \neg[((\neg p \wedge \neg q) \vee (q \wedge \neg q)) \vee ((\neg p \wedge p)) \vee (q \wedge p))]$,
by distributive law
$\equiv \neg[((\neg p \wedge \neg q) \vee F) \vee ((F \vee (q \wedge p))]$, by complement law
$\equiv \neg[(\neg p \wedge \neg q) \vee (q \wedge p)]$, by identity law
$\equiv \neg[\neg(p \vee q) \vee (q \wedge p)]$, by De Morgan's law
$\equiv (p \vee q) \wedge \neg(q \wedge p)$, by De Morgan's law (1)
$\equiv (p \vee q) \wedge (\neg q \vee \neg p)$, by De Morgan's law
$\equiv ((p \vee q) \wedge \neg q)) \vee ((p \vee q) \wedge \neg p))$, by distributive law
$\equiv ((p \wedge \neg q) \vee (q \wedge \neg q)) \vee ((p \wedge \neg p) \vee (q \wedge \neg p))$, by distributive law

$\equiv ((p \wedge \neg q) \vee F) \vee ((F \vee (q \wedge \neg p))$, by complement law

$\equiv (p \wedge \neg q) \vee (q \wedge \neg p)$, by identity law

$\equiv (p \wedge \neg q) \vee (\neg p \wedge q)$, by commutative law

From (1) and (2), the result follows. (2)

Example 1.5 Without constructing the truth tables, prove the following

(i) $\neg p \rightarrow (q \rightarrow r) \equiv q \rightarrow (p \vee r)$

(ii) $p \rightarrow (q \rightarrow r) \equiv p \rightarrow (\neg q \vee r) \equiv (p \wedge q) \rightarrow r$

(iii) $((p \vee q) \wedge \neg(\neg p \wedge (\neg q \vee \neg r))) \vee (\neg p \wedge \neg q) \vee (\neg p \wedge \neg r)$ is a tautology.

(i) $\neg p \rightarrow (q \rightarrow r) \equiv p \vee (q \rightarrow r)$, from Table 1.10

$\equiv p \vee (\neg q \vee r)$, from Table 1.10

$\equiv (p \vee \neg q) \vee r$, by associative law

$\equiv (\neg q \vee p) \vee r$, by commutative law

$\equiv \neg q \vee (p \vee r)$, by associative law

$\equiv q \rightarrow (p \vee r)$, from Table 1.10.

(ii) $p \rightarrow (q \rightarrow r) \equiv p \rightarrow (\neg q \vee r)$, from Table 1.10 (1)

Now $p \rightarrow (\neg q \vee r) \equiv \neg p \vee (\neg q \vee r)$, from Table 1.10

$\equiv (\neg p \vee \neg q) \vee r$, by associative law

$\equiv \neg(p \wedge q) \vee r$, by De Morgan's law

$\equiv (p \wedge q) \rightarrow r$ (2)

(iii) $((p \vee q) \wedge \neg(\neg p \wedge (\neg q \vee \neg r))) \vee (\neg p \wedge \neg q) \vee (\neg p \wedge \neg r)$

$\equiv ((p \vee q) \wedge \neg(\neg p \wedge \neg(q \wedge r))) \vee \neg(p \vee q) \vee \neg(p \vee r)$,

by De Morgan's law

$\equiv ((p \vee q) \wedge (p \vee (q \wedge r))) \vee \neg(p \vee q) \vee \neg(p \vee r)$,

by De Morgan's law

$\equiv ((p \vee q) \wedge [(p \vee q) \wedge (p \vee r)]) \vee [\neg(p \vee q) \vee \neg(p \vee r)]$,

by distributive law

$\equiv [(p \vee q) \wedge (p \vee r)] \vee \neg[(p \vee q) \wedge (p \vee r)]$,

by idempotent and De Morgan's laws

The final statement is in the form of $p \vee \neg p$.

$\therefore$ L.H.S. $\equiv$ T

Hence the given statement is tautology.

Example 1.6 Prove the following equivalences by proving the equivalences of the duals:

(i) $\neg((\neg p \wedge q) \vee (\neg p \wedge \neg q)) \vee (p \wedge q) \equiv p$

(ii) $(p \vee q) \rightarrow r \equiv (p \rightarrow r) \wedge (q \wedge r)$

(iii) $(p \wedge (p \leftrightarrow q)) \rightarrow q \equiv T$

(i) The dual of the given equivalence is

$$\neg((\neg p \vee q) \wedge (\neg p \vee \neg q)) \wedge (p \vee q) \equiv p$$

Let us now prove the dual equivalence.

L.H.S. $\equiv \neg(\neg p \vee (q \wedge \neg q)) \wedge (p \vee q)$, by distribution law

$\equiv \neg(\neg p \vee F) \wedge (p \vee q)$, by complement law

$\equiv \neg(\neg p) \wedge (p \vee q)$, by identity law

$\equiv p \wedge (p \vee q)$

$\equiv p$, by absorption law

(ii) $(p \vee q) \rightarrow r \equiv (p \rightarrow r) \wedge (q \rightarrow r)$

i.e., $\neg(p \vee q) \vee r \equiv (\neg p \vee r) \wedge (\neg q \vee r)$

Dual of this equivalence is

$$\neg(p \wedge q) \wedge r \equiv (\neg p \wedge r) \vee (\neg q \wedge r)$$

L.H.S $\equiv (\neg p \vee \neg q) \wedge r$, by De Morgan's law

$\equiv (\neg p \wedge r) \vee (\neg q \wedge r)$, by distributive law

$\equiv$ R.H.S.

(iii) $(p \wedge (p \leftrightarrow q)) \rightarrow q \equiv T$

i.e. $p \wedge ((p \rightarrow q) \wedge (q \rightarrow p)) \rightarrow q \equiv T$, from Table 1.11

i.e. $p \wedge ((\neg p \vee q) \wedge (\neg q \vee p)) \rightarrow q \equiv T$

i.e. $\neg(p \wedge ((\neg p \vee q) \wedge (\neg q \vee p))) \vee q \equiv T$

Dual of this equivalence is

$$\neg(p \vee ((\neg p \wedge q) \vee (\neg q \wedge p))) \wedge q \equiv F$$

L.H.S. $\equiv \neg[(p \vee (\neg p \wedge q)) \vee (\neg q \wedge p)] \wedge q$, by associative law

$\equiv \neg[(\mathrm{T} \wedge (p \vee q)) \vee (\neg q \wedge p)] \wedge q$, by distributive and complement laws

$\equiv \neg[(p \vee q) \vee (\neg q \wedge p)] \wedge q$, by identity law

$\equiv \neg[((p \vee q) \vee \neg q) \wedge ((p \vee q) \vee p)] \wedge q$, by distributive law

$\equiv \neg[(p \vee T) \wedge (p \vee q)] \wedge q$, by idempotent and complement laws.

$\equiv \neg[T \wedge (p \vee q)] \wedge q$, by dominant law

$\equiv \neg[p \vee q] \wedge q$, by identity law

$\equiv (\neg p \wedge \neg q) \wedge q$, by De Morgan's law

$\equiv (\neg p \wedge F)$, by complement law

$\equiv F$, by dominant law.